

Course Information

Degree: UG

Program: BCA

Course Name: Software Engineering

Course Credits: 3

Semester: 3rd

Course Category: Core

Type of Course: Lecture/ Experiential

New Course: Yes

Learning Hours

Activity	Hours per Week
Lecture	3
Tutorial	0
Practical	0
Experiential Learning	3

Course Contribution

Category	Percentage Contribution
Employment Enhancement	50%
Entrepreneurship Skills	20%
Skill Development	30%

Course Relevance

Level	Percentage Relevance
Global	40%
National	30%
Regional	20%
Local	10%

Course Outcomes (COs)

CO#	Course Outcome	Bloom's Taxonomy Level
CO1	Describe the evolution of software development life cycle models and their relevance in modern software engineering.	BL-2
CO2	Apply software requirements engineering principles and project management principles to enhance risk assessment, scheduling, and estimation in software development.	BL-3

C03	Construct software design models by applying design concepts, architectural design, data design, and structural modeling using UML diagrams.	BL-3
C04	Implement software testing strategies, quality metrics, and debugging techniques for defect identification.	BL-3
C05	Assess software quality assurance processes and release management strategies for software maintenance and sustenance.	BL-4

Module	Course Content	Tools Used / Assessment and Activity	Bloom's Level	Mapping COs	Percentage Change in Module
Module 1: Software Process and Agile Development	The evolving role of software, changing nature of software		BL-2	CO1	NEW
	Layered technology, a process framework		BL-3	CO3	NEW
	Process models: The waterfall model		BL-2	CO1	NEW
	Incremental process models		BL-3	CO5	NEW
	Evolutionary process models		BL-4	CO4	NEW
	The unified process		BL-2	CO3	NEW
	Agile software development: Agility Principles		BL-3	CO1	NEW
	Agile methods, Plan-driven and agile development		BL-4	CO2	NEW
	Extreme programming, Scrum, A Tool Set for the Agile Process		BL-3	CO5	NEW
Module 2: Software Requirements and Risk Management	Software Requirements Engineering: Functional and non-functional requirements		BL-2	CO4	NEW
	Software requirements document,		BL-3	CO2	NEW

	Requirements specification				
	Requirements engineering processes		BL-4	CO1	NEW
	Requirements elicitation and analysis		BL-3	CO5	NEW
	Requirements validation, Requirements management		BL-4	CO3	NEW
	Risk management: Reactive Vs proactive risk strategies		BL-2	CO2	NEW
	Software risks, risk identification, risk projection, risk refinement		BL-3	CO4	NEW
	RMMM, RMMM plan		BL-4	CO5	NEW
	Project planning- Software pricing, Plan-driven development, Project scheduling, Agile planning, Estimation techniques		BL-3	CO3	NEW
Module 3: Software Design and UML Modeling	Design process and design quality		BL-2	CO1	NEW
	Design concepts, the design model		BL-3	CO4	NEW
	Software architecture		BL-4	CO5	NEW
	Data design, architectural design		BL-3	CO2	NEW
	Basic structural modeling		BL-2	CO3	NEW
	Class diagrams, sequence diagrams		BL-3	CO5	NEW
	Collaboration diagrams, use case diagrams		BL-4	CO1	NEW

	Component diagrams		BL-3	CO4	NEW
	Summary and Review of Design Concepts		BL-2	CO3	NEW
Module 4: Software Testing and Product Metrics	Testing Strategies: A strategic approach to software testing		BL-2	CO2	NEW
	Test strategies for conventional software		BL-3	CO3	NEW
	Black-box and white-box testing		BL-4	CO1	NEW
	Validation testing, system testing		BL-3	CO5	NEW
	The art of debugging		BL-2	CO4	NEW
	Product metrics: Software quality, metrics for analysis model		BL-3	CO3	NEW
	Metrics for design model, metrics for source code		BL-4	CO2	NEW
	Metrics for testing, metrics for maintenance		BL-3	CO1	NEW
	Summary and Review of Testing Strategies		BL-2	CO4	NEW
Module 5: Quality	Quality Management: Quality concepts		BL-2	CO4	NEW
	Software quality assurance		BL-3	CO1	NEW
	Software reviews, formal technical reviews		BL-4	CO2	NEW
	Statistical software quality assurance		BL-3	CO5	NEW
	Software reliability		BL-2	CO3	NEW
	Release Management: Release planning,		BL-3	CO2	NEW

	development and build plans				
	Release strategies, risk management, and post-deployment monitoring		BL-4	CO1	NEW
	Product sustenance: Maintenance, updates		BL-3	CO5	NEW
	End of life, migration strategies		BL-2	CO3	NEW

Textbooks:

1	Software Engineering A practitioner's Approach, 8th edition, Roger S Pressman, Bruce R. Maxim. McGraw Hill Education, 2015.
2	Software Engineering, Ian Somerville, 9th edition, Pearson education

CO-PO/PSO Mapping

CO	P01	P02	P03	P04	P05	P06	P07	P08	P09
CO1	2	2				1			
CO2		3	2		1	3	1		
CO3			3	2	2				2
CO4		2		3	2		3	1	
CO5				2		2	3	2	2